

AN EXPLICIT NONDEGENERATE T_4 -CONFIGURATION IN THE CONSTITUTIVE SET OF A GENUINELY NONLINEAR 2×2 SYSTEM OF CONSERVATION LAWS

ABSTRACT. We give a constructive example of a smooth flux for a one-dimensional 2×2 system of conservation laws which is strictly hyperbolic and genuinely nonlinear on a convex open state domain and which admits a strictly convex entropy. The associated entropy-augmented constitutive set in $\mathbb{R}^{3 \times 2}$ contains an ordered T_4 -configuration. The configuration is nondegenerate in the stronger sense that no two of its four wells are rank-one connected. All matrices and all finite interpolation data are rational and explicit. The flux is obtained from a scalar potential. A one-dimensional Hermite interpolant places the four prescribed wells, while a transverse smooth connector enforces genuine nonlinearity of both characteristic families. We include the exact rank calculations, the full connector construction, and the passage from the central state segment to a convex open strip.

1. INTRODUCTION AND MAIN RESULT

Let $\mathcal{V} \subset \mathbb{R}^2$ be a nonempty convex open set and consider the system of conservation laws

$$U_t + F(U)_x = 0, \quad U: \mathbb{R} \times [0, \infty) \longrightarrow \mathcal{V}, \quad (1)$$

with a smooth flux $F = (F_1, F_2): \mathcal{V} \rightarrow \mathbb{R}^2$. We regard differentials of scalar functions as row covectors. Thus a pair of smooth scalar functions (η, q) is an entropy–entropy-flux pair for equation (1) when

$$Dq(U) = D\eta(U) DF(U), \quad U \in \mathcal{V}. \quad (2)$$

The entropy is strictly convex when $D^2\eta(U)$ is positive definite for every $U \in \mathcal{V}$.

The system is strictly hyperbolic when $DF(U)$ has two distinct real eigenvalues

$$\lambda_1(U) < \lambda_2(U), \quad U \in \mathcal{V}. \quad (3)$$

If $r_k(U)$ is a corresponding right eigenvector, the k th characteristic field is genuinely nonlinear when

$$D\lambda_k(U)[r_k(U)] \neq 0, \quad U \in \mathcal{V}, \quad k = 1, 2. \quad (4)$$

For $U = (u_1, u_2) \in \mathcal{V}$, define

$$G_{F,\eta,q}(U) := \begin{pmatrix} u_1 & -F_1(U) \\ u_2 & -F_2(U) \\ \eta(U) & -q(U) \end{pmatrix}. \quad (5)$$

The entropy-augmented constitutive set is

$$K_{F,\eta,q} := G_{F,\eta,q}(\mathcal{V}) \subset \mathbb{R}^{3 \times 2}. \quad (6)$$

For a fixed positive integer N , it is useful to distinguish the single-well set in equation (6) from its N -fold product:

$$\mathcal{K}_{F,\eta,q}^{(N)} := (K_{F,\eta,q})^N \subset (\mathbb{R}^{3 \times 2})^N. \quad (7)$$

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Our main result is the following explicit affirmative construction.

Theorem 1.1 (A nondegenerate constitutive T_4). *There exist a number $\delta > 0$, a convex open rectangle*

$$\mathcal{V} = (-8, 8) \times (-\delta, \delta), \quad (8)$$

smooth functions

$$F: \mathcal{V} \rightarrow \mathbb{R}^2, \quad \eta, q: \mathcal{V} \rightarrow \mathbb{R}, \quad (9)$$

and four states $U_1, \dots, U_4 \in \mathcal{V}$ with the following properties:

- (1) $D^2\eta = I_2$, so η is uniformly strictly convex;
- (2) (η, q) satisfies equation (2);
- (3) the system equation (1) is strictly hyperbolic throughout $\mathcal{V}$;
- (4) both characteristic fields are genuinely nonlinear throughout $\mathcal{V}$;
- (5) the ordered tuple

$$(G_{F,\eta,q}(U_1), \dots, G_{F,\eta,q}(U_4)) \in \mathcal{K}_{F,\eta,q}^{(4)} \quad (10)$$

is an ordered T_4 -configuration;

- (6) *no two distinct members of this configuration are rank-one connected.*

In fact, one may take

$$U_1 = (-7, 0), \quad U_2 = (-1, 0), \quad U_3 = (1, 0), \quad U_4 = (7, 0), \quad (11)$$

and the flux may be chosen to be a gradient, $F = \nabla\Phi$, for a smooth scalar potential Φ .

The proof separates into two parts. First, we give an exact rational T_4 -configuration and determine the finite jets that a potential must realize at the states in equation (11). Second, we construct a smooth potential realizing those jets while preserving strict hyperbolicity and genuine nonlinearity on a thin convex strip.

2. ORDERED T_N -CONFIGURATIONS AND POTENTIAL FLUXES

Set

$$\mathbb{M} := \mathbb{R}^{3 \times 2}, \quad \mathcal{R}_1 := \{C \in \mathbb{M} : \text{rank } C = 1\}. \quad (12)$$

Thus $\mathcal{R}_1$ is the manifold of nonzero rank-one matrices in $\mathbb{M}$.

Definition 2.1 (Ordered T_N -configuration). Fix $N \geq 4$. An ordered N -tuple

$$\mathbf{X} = (X_1, \dots, X_N) \in \mathbb{M}^N \quad (13)$$

is an ordered T_N -configuration if there exist

$$P \in \mathbb{M}, \quad C_i \in \mathcal{R}_1, \quad \kappa_i > 1, \quad i = 1, \dots, N, \quad (14)$$

such that

$$\sum_{i=1}^N C_i = 0 \quad (15)$$

and

$$X_i = P + \sum_{j=1}^{i-1} C_j + \kappa_i C_i, \quad i = 1, \dots, N. \quad (16)$$

The tuple

$$(P, C_1, \dots, C_N, \kappa_1, \dots, \kappa_N) \quad (17)$$

is called an ordered T_N -parameterization of $\mathbf{X}$.

Definition 2.2 (Nondegeneracy). An ordered T_N -configuration in $\mathbb{R}^{3 \times 2}$ is called nondegenerate if

$$\text{rank}(X_i - X_j) = 2, \quad i \neq j. \quad (18)$$

Because a 3×2 matrix has rank at most two, condition equation (18) is precisely the assertion that no two distinct wells are rank-one connected.

The gradient-flux ansatz makes entropy compatibility automatic.

Lemma 2.3 (Potential-generated entropy pair). *Let Φ be smooth on an open subset of $\mathbb{R}^2$ and define*

$$F(U) := \nabla \Phi(U), \quad \eta(U) := \frac{1}{2}|U|^2, \quad q(U) := U \cdot \nabla \Phi(U) - \Phi(U). \quad (19)$$

Then $D^2\eta = I_2$ and (η, q) satisfies equation (2).

Proof. For $v \in \mathbb{R}^2$, the product rule gives

$$Dq(U)[v] = v \cdot \nabla \Phi(U) + U \cdot D^2\Phi(U)v - D\Phi(U)[v] \quad (20)$$

$$= U \cdot D^2\Phi(U)v, \quad (21)$$

because $D\Phi(U)[v] = \nabla \Phi(U) \cdot v$. Since

$$D\eta(U) = U^\top, \quad DF(U) = D^2\Phi(U), \quad (22)$$

we obtain $Dq(U) = D\eta(U)DF(U)$. The identity $D^2\eta = I_2$ is immediate. $\square$

We shall also use the following standard eigenvalue differentiation formula, for which we include the proof to fix all conventions.

Lemma 2.4 (Directional derivative of a simple Hessian eigenvalue). *Let $\Phi \in C^3(\Omega)$, let $D^2\Phi(U)$ have a simple eigenvalue $\lambda_k(U)$, and let $r_k(U)$ be a differentiable unit eigenvector. Then, for every $v \in \mathbb{R}^2$,*

$$D\lambda_k(U)[v] = D^3\Phi(U)[v, r_k(U), r_k(U)]. \quad (23)$$

In particular,

$$D\lambda_k(U)[r_k(U)] = D^3\Phi(U)[r_k(U), r_k(U), r_k(U)]. \quad (24)$$

Proof. Differentiate

$$D^2\Phi(U)r_k(U) = \lambda_k(U)r_k(U) \quad (25)$$

in the direction v and take the Euclidean inner product with $r_k(U)$. The terms containing $Dr_k(U)[v]$ cancel because $D^2\Phi(U)$ is symmetric and $D^2\Phi(U)r_k(U) = \lambda_k(U)r_k(U)$. Since $|r_k(U)| = 1$, the remaining identity is exactly equation (23). Taking $v = r_k(U)$ gives equation (24). $\square$

3. AN EXACT RATIONAL T_4 CERTIFICATE

We now give the algebraic core of the construction. Take

$$\kappa_1 = \kappa_2 = \kappa_3 = \kappa_4 = 2 \quad (26)$$

and

$$P = \begin{pmatrix} \frac{17}{5} & 0 \\ 0 & 0 \\ \frac{149}{10} & 0 \end{pmatrix}. \quad (27)$$

Define the four increments by

$$C_1 = \begin{pmatrix} -\frac{26}{5} & -\frac{26}{5} \\ 0 & 0 \\ \frac{24}{5} & \frac{24}{5} \end{pmatrix}, \quad C_2 = \begin{pmatrix} \frac{2}{5} & -\frac{4}{5} \\ 0 & 0 \\ -\frac{48}{5} & \frac{96}{5} \end{pmatrix}, \quad (28)$$

$$C_3 = \begin{pmatrix} \frac{6}{5} & 6 \\ 0 & 0 \\ -\frac{24}{5} & -24 \end{pmatrix}, \quad C_4 = \begin{pmatrix} \frac{18}{5} & 0 \\ 0 & 0 \\ \frac{48}{5} & 0 \end{pmatrix}. \quad (29)$$

Their rank-one structure is transparent from the outer-product factorizations

$$C_1 = \begin{pmatrix} -\frac{26}{5} \\ 0 \\ \frac{24}{5} \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad C_2 = \begin{pmatrix} \frac{2}{5} \\ 0 \\ -\frac{48}{5} \end{pmatrix} \begin{pmatrix} 1 & -2 \end{pmatrix}, \quad (30)$$

$$C_3 = \begin{pmatrix} \frac{6}{5} \\ 0 \\ -\frac{24}{5} \end{pmatrix} \begin{pmatrix} 1 & 5 \end{pmatrix}, \quad C_4 = \begin{pmatrix} \frac{18}{5} \\ 0 \\ \frac{48}{5} \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}. \quad (31)$$

Every column factor and every row factor in equations (30) and (31) is nonzero. Hence $C_i \in \mathcal{R}_1$ for every i . Direct summation gives

$$\sum_{i=1}^4 C_i = \begin{pmatrix} \frac{-26+2+6+18}{5} & \frac{-26-4+30}{5} \\ 0 & 0 \\ \frac{24-48-24+48}{5} & \frac{24+96-120}{5} \end{pmatrix} = 0_{3 \times 2}, \quad (32)$$

which verifies equation (15).

Using equation (16) with the order displayed above and $\kappa_i = 2$, we obtain

$$X_1 = P + 2C_1 = \begin{pmatrix} -7 & -\frac{52}{5} \\ 0 & 0 \\ \frac{49}{2} & \frac{48}{5} \end{pmatrix}, \quad X_2 = P + C_1 + 2C_2 = \begin{pmatrix} -1 & -\frac{34}{5} \\ 0 & 0 \\ \frac{1}{2} & \frac{216}{5} \end{pmatrix}, \quad (33)$$

$$X_3 = P + C_1 + C_2 + 2C_3 = \begin{pmatrix} 1 & 6 \\ 0 & 0 \\ \frac{1}{2} & -24 \end{pmatrix}, \quad X_4 = P + C_1 + C_2 + C_3 + 2C_4 = \begin{pmatrix} 7 & 0 \\ 0 & 0 \\ \frac{49}{2} & 0 \end{pmatrix}. \quad (34)$$

Thus (X_1, X_2, X_3, X_4) is an ordered T_4 -configuration. It remains at this algebraic stage to verify the stronger pairwise nondegeneracy property.

For $1 \leq i < j \leq 4$, let

$$\Delta_{ij} := \det \left((X_i - X_j)_{\{1,3\} \times \{1,2\}} \right), \quad (35)$$

where we retain the first and third rows and both columns. The first three submatrices and determinants are

(i, j)	$(X_i - X_j)_{\{1,3\} \times \{1,2\}}$	Δ_{ij}
(1, 2)	$\begin{pmatrix} -6 & -\frac{18}{5} \\ 24 & -\frac{168}{5} \end{pmatrix}$	288
(1, 3)	$\begin{pmatrix} -8 & -\frac{82}{5} \\ 24 & \frac{168}{5} \end{pmatrix}$	$\frac{624}{5}$
(1, 4)	$\begin{pmatrix} -14 & -\frac{52}{5} \\ 0 & \frac{48}{5} \end{pmatrix}$	$-\frac{672}{5}$

(36)

and the remaining three are

$$\begin{array}{c|c|c}
 (i, j) & (X_i - X_j)_{\{1,3\} \times \{1,2\}} & \Delta_{ij} \\
 \hline
 (2, 3) & \begin{pmatrix} -2 & -\frac{64}{5} \\ 0 & \frac{336}{5} \end{pmatrix} & -\frac{672}{5} \\
 (2, 4) & \begin{pmatrix} -8 & -\frac{34}{5} \\ -24 & \frac{216}{5} \end{pmatrix} & -\frac{2544}{5} \\
 (3, 4) & \begin{pmatrix} -6 & 6 \\ -24 & -24 \end{pmatrix} & 288
 \end{array} \tag{37}$$

Every determinant in equations (36) and (37) is nonzero. Therefore $\text{rank}(X_i - X_j) \geq 2$ for $i \neq j$. Since these are 3×2 matrices, their ranks are at most two, and hence

$$\text{rank}(X_i - X_j) = 2, \quad i \neq j. \tag{38}$$

4. REDUCTION TO FINITE POTENTIAL JETS

We now realize the matrices in equations (33) and (34) as constitutive wells. Use the potential ansatz from lemma 2.3. Write a state as $U = (x, y)$ and let

$$\eta(x, y) = \frac{x^2 + y^2}{2}. \tag{39}$$

Suppose that $\Phi_y(x, 0) = 0$ and set

$$\psi(x) := \Phi(x, 0). \tag{40}$$

At a state $(x, 0)$ we then have

$$F(x, 0) = (\psi'(x), 0), \quad q(x, 0) = x\psi'(x) - \psi(x), \tag{41}$$

and consequently

$$G_{F, \eta, q}(x, 0) = \begin{pmatrix} x & -\psi'(x) \\ 0 & 0 \\ \frac{x^2}{2} & \psi(x) - x\psi'(x) \end{pmatrix}. \tag{42}$$

Let

$$\xi_1 = -7, \quad \xi_2 = -1, \quad \xi_3 = 1, \quad \xi_4 = 7. \tag{43}$$

Comparing equation (42) with equations (33) and (34), one sees that it is enough to impose $\Phi_y(\xi_i, 0) = 0$ together with the following scalar Hermite data:

$$\begin{array}{c|cccc}
 x & -7 & -1 & 1 & 7 \\
 \hline
 \psi(x) & -\frac{316}{5} & \frac{182}{5} & -30 & 0 \\
 \psi'(x) & \frac{52}{5} & \frac{34}{5} & -6 & 0
 \end{array} . \tag{44}$$

For example, at $x = -7$ the upper-right entry of X_1 forces $\psi'(-7) = 52/5$, and then its lower-right entry forces

$$\psi(-7) - (-7)\psi'(-7) = \frac{48}{5}, \tag{45}$$

which gives $\psi(-7) = -316/5$. The other three columns of equation (44) follow by the same calculation.

5. THE ONE-DIMENSIONAL INTERPOLANT AND THE TRANSVERSE CONNECTOR

5.1. **The Hermite polynomial.** Define the degree-seven polynomial

$$\begin{aligned} \psi_H(x) := & \frac{33263}{5120} - \frac{4085033}{80640}x - \frac{121651}{35840}x^2 + \frac{23868737}{1317120}x^3 \\ & + \frac{3531}{35840}x^4 - \frac{176725}{263424}x^5 - \frac{33}{35840}x^6 + \frac{26129}{3951360}x^7. \end{aligned} \quad (46)$$

This is the unique polynomial of degree at most seven satisfying the eight conditions in equation (44). Introduce

$$R(x) := (x^2 - 1)(x^2 - 49) \quad (47)$$

and set

$$\psi(x) := \psi_H(x) - xR(x)^2. \quad (48)$$

At each $\xi_i \in \{-7, -1, 1, 7\}$ one has $R(\xi_i) = 0$. Therefore

$$(xR(x)^2)|_{x=\xi_i} = 0 \quad (49)$$

and, since

$$\frac{d}{dx}(xR(x)^2) = R(x)^2 + 2xR(x)R'(x), \quad (50)$$

its first derivative also vanishes at every ξ_i . Hence ψ and ψ_H have the same values and first derivatives at the four nodes. A direct exact evaluation gives

x	-7	-1	1	7
$\psi_H(x) = \psi(x)$	$-\frac{316}{5}$	$\frac{182}{5}$	-30	0
$\psi'_H(x) = \psi'(x)$	$\frac{52}{5}$	$\frac{34}{5}$	-6	0

(51)

which verifies equation (44).

The correction in equation (48) is chosen to create regions in which the third derivative is positive without altering the required jets. Indeed, exact differentiation yields

$$\begin{aligned} \psi'''(-5) &= \frac{455187556553}{329280} > 0, & \psi'''(0) &= \frac{6477756737}{219520} > 0, \\ \psi'''(5) &= \frac{455186246783}{329280} > 0. \end{aligned} \quad (52)$$

By continuity, there are pairwise disjoint open intervals

$$J_{-5} \Subset (-7, -1), \quad J_0 \Subset (-1, 1), \quad J_5 \Subset (1, 7), \quad (53)$$

centered at $-5, 0, 5$, respectively, such that

$$\overline{J_{-5}} \cup \overline{J_0} \cup \overline{J_5} \subset \{x \in \mathbb{R} : \psi'''(x) > 0\}. \quad (54)$$

5.2. **A smooth connector with four prescribed zeros.** For completeness, the bump functions used below may be chosen from the standard profile

$$\vartheta(t) := \begin{cases} \exp(-1/(1-t^2)), & |t| < 1, \\ 0, & |t| \geq 1. \end{cases} \quad (55)$$

Choose $\rho_a > 0$ so small that $[a - \rho_a, a + \rho_a] \subset J_a$ and take

$$\alpha_a(x) := \vartheta\left(\frac{x-a}{\rho_a}\right), \quad a \in \{-5, 0, 5\}. \quad (56)$$

Thus $\alpha_a \in C_c^\infty(J_a)$ is nonnegative and nonzero. Normalize these bumps by

$$\beta_{-5} := \frac{6\alpha_{-5}}{\int_{\mathbb{R}} \alpha_{-5}(t) dt}, \quad \beta_0 := \frac{2\alpha_0}{\int_{\mathbb{R}} \alpha_0(t) dt}, \quad \beta_5 := \frac{6\alpha_5}{\int_{\mathbb{R}} \alpha_5(t) dt}. \quad (57)$$

Define

$$g(x) := -1 + \beta_{-5}(x) + \beta_0(x) + \beta_5(x), \quad h(x) := \int_{-7}^x g(t) dt. \quad (58)$$

Since the three supports lie in disjoint intervals, we have

$$\int_{-7}^{-1} g(t) dt = -6 + \int_{\mathbb{R}} \beta_{-5}(t) dt = 0, \quad (59)$$

$$\int_{-1}^1 g(t) dt = -2 + \int_{\mathbb{R}} \beta_0(t) dt = 0, \quad (60)$$

$$\int_1^7 g(t) dt = -6 + \int_{\mathbb{R}} \beta_5(t) dt = 0. \quad (61)$$

It follows successively from equations (58) to (61) that

$$h(-7) = h(-1) = h(1) = h(7) = 0. \quad (62)$$

Let

$$I := [-8, 8], \quad E := \{x \in I : g(x) \geq 0\}. \quad (63)$$

Outside the union of the three bump supports, $g = -1$. Consequently,

$$E \subset \bigcup_{a \in \{-5, 0, 5\}} \text{supp } \beta_a \subseteq \{x \in \mathbb{R} : \psi'''(x) > 0\}. \quad (64)$$

The set E is compact. It is also nonempty: on each of the intervals of length 6, 2, 6, respectively, the corresponding β_a has integral equal to that length and vanishes near both endpoints, so it must exceed 1 somewhere. In particular,

$$\mu := \min_{x \in E} \psi'''(x) > 0. \quad (65)$$

5.3. Choice of the potential on the central segment. Choose the concrete constant

$$L := 2 + \|\psi''\|_{L^\infty(I)} \quad (66)$$

and consider potentials of the form

$$\Phi(x, y) = \psi(x) + \varepsilon h(x)y + \frac{L}{2}y^2 + \frac{b(x)}{6}y^3, \quad (67)$$

where $\varepsilon > 0$ will be chosen small and the smooth function b will be fixed after that choice. On $y = 0$,

$$D^2\Phi(x, 0) = \begin{pmatrix} \psi''(x) & \varepsilon g(x) \\ \varepsilon g(x) & L \end{pmatrix}. \quad (68)$$

Set

$$d(x) := L - \psi''(x), \quad d_0 := \min_{x \in I} d(x) \geq 2. \quad (69)$$

The two eigenvalues of equation (68) are

$$\lambda_{1,2}(x, 0) = \frac{\psi''(x) + L \mp \sqrt{d(x)^2 + 4\varepsilon^2 g(x)^2}}{2}. \quad (70)$$

Their gap is therefore

$$\lambda_2(x, 0) - \lambda_1(x, 0) = \sqrt{d(x)^2 + 4\varepsilon^2 g(x)^2} \geq d_0 \geq 2. \quad (71)$$

We next fix orientations of the two eigenvectors. Define

$$\Delta(x) := \sqrt{d(x)^2 + 4\varepsilon^2 g(x)^2}, \quad \tau(x) := -\frac{2\varepsilon g(x)}{\Delta(x) + d(x)}, \quad (72)$$

and

$$c(x) := \frac{1}{\sqrt{1 + \tau(x)^2}}, \quad \sigma(x) := \tau(x)c(x). \quad (73)$$

Then

$$r_1(x, 0) := (c(x), \sigma(x)), \quad r_2(x, 0) := (-\sigma(x), c(x)) \quad (74)$$

are unit eigenvectors corresponding to the lower and upper eigenvalues in equation (70). Indeed,

$$\frac{\lambda_1(x, 0) - \psi''(x)}{\varepsilon g(x)} = -\frac{2\varepsilon g(x)}{\Delta(x) + d(x)} = \tau(x) \quad (75)$$

when $g(x) \neq 0$, and the formula extends continuously across the zeros of g . The second vector is the positively oriented orthogonal complement of the first.

Because $\Delta \geq d > 0$,

$$|\tau(x)| = \frac{2\varepsilon|g(x)|}{\Delta(x) + d(x)} \leq \frac{\varepsilon|g(x)|}{d(x)} \leq \frac{\varepsilon\|g\|_{L^\infty(I)}}{d_0}. \quad (76)$$

Moreover, $|\tau(x)| < 1$, so

$$c(x) > \frac{1}{\sqrt{2}}, \quad \text{sgn } \sigma(x) = -\text{sgn } g(x), \quad (77)$$

where the sign equality is understood to give $\sigma = 0$ when $g = 0$. It follows from equation (76) that, uniformly on I ,

$$\sigma \longrightarrow 0, \quad c \longrightarrow 1 \quad \text{as } \varepsilon \longrightarrow 0. \quad (78)$$

Along $y = 0$, the only possibly nonzero components of $D^3\Phi$ are

$$\Phi_{xxx} = \psi''', \quad \Phi_{xxy} = \varepsilon g', \quad \Phi_{xyy} = 0, \quad \Phi_{yyy} = b. \quad (79)$$

Consequently,

$$D^3\Phi[r_1, r_1, r_1] = A_1 + b\sigma^3, \quad (80)$$

$$D^3\Phi[r_2, r_2, r_2] = A_2 + bc^3, \quad (81)$$

where all quantities in equations (80) and (81) are evaluated at $(x, 0)$ and

$$A_1(x) := \psi'''(x)c(x)^3 + 3\varepsilon g'(x)c(x)^2\sigma(x), \quad (82)$$

$$A_2(x) := -\psi'''(x)\sigma(x)^3 + 3\varepsilon g'(x)\sigma(x)^2c(x). \quad (83)$$

The uniform limits in equation (78) imply

$$A_1 + \sigma^3 \longrightarrow \psi''' \quad \text{uniformly on } I, \quad A_2 + c^3 \longrightarrow 1 \quad \text{uniformly on } I. \quad (84)$$

Choose $\varepsilon > 0$ sufficiently small that

$$A_1(x) + \sigma(x)^3 \geq \frac{\mu}{2} \quad \text{for every } x \in E, \quad (85)$$

and

$$A_2(x) + c(x)^3 \geq \frac{1}{2} \quad \text{for every } x \in I. \quad (86)$$

Such a choice is possible by equations (65) and (84).

It remains to choose b . Since $g \geq 0$ on E , we have $\sigma \leq 0$ there by equation (77). Thus equation (85) implies

$$A_1 \geq \frac{\mu}{2} - \sigma^3 > 0 \quad \text{on } E. \quad (87)$$

Define the bad set

$$\mathcal{B} := \{x \in I : A_1(x) \leq 0\}. \quad (88)$$

It is a closed subset of the compact interval I , hence compact, and equation (87) shows that

$$\mathcal{B} \subseteq \{x \in \mathbb{R} : g(x) < 0\}. \quad (89)$$

The notation $\subseteq$ in equation (89) is justified as follows: $\mathcal{B}$ is compact, the set $\{g < 0\}$ is open, and $\mathcal{B} \cap \{g \geq 0\} = \emptyset$.

If $\mathcal{B} = \emptyset$, take $\chi = 0$ and $M = 0$. If $\mathcal{B} \neq \emptyset$, the smooth cutoff lemma gives a function

$$\chi \in C_c^\infty(\{x \in \mathbb{R} : g(x) < 0\}), \quad 0 \leq \chi \leq 1, \quad \chi = 1 \text{ on } \mathcal{B}. \quad (90)$$

On $\mathcal{B}$ one has $g < 0$ and hence $\sigma > 0$. Compactness gives $\min_{\mathcal{B}} \sigma > 0$. Choose

$$M > \max_{x \in \mathcal{B}} \left(\frac{-A_1(x) - \sigma(x)^3}{\sigma(x)^3} \right)_+ \quad (91)$$

and define

$$b(x) := 1 + M\chi(x). \quad (92)$$

Here $z_+ := \max\{z, 0\}$. In the case $\mathcal{B} = \emptyset$, condition equation (91) is omitted and the choice $M = 0$ is retained.

We now verify the first cubic contraction on all of I . There are three cases.

(1) If $g(x) \geq 0$, then $x \in E$ and $\chi(x) = 0$, because $\text{supp } \chi \subset \{g < 0\}$. Hence $b(x) = 1$ and

$$A_1(x) + b(x)\sigma(x)^3 = A_1(x) + \sigma(x)^3 \geq \frac{\mu}{2} > 0. \quad (93)$$

(2) If $g(x) < 0$ and $A_1(x) > 0$, then $\sigma(x) > 0$ and $b(x) \geq 1$, so

$$A_1(x) + b(x)\sigma(x)^3 > 0. \quad (94)$$

(3) If $g(x) < 0$ and $A_1(x) \leq 0$, then $x \in \mathcal{B}$, $\chi(x) = 1$, and the choice equation (91) gives

$$A_1(x) + (1 + M)\sigma(x)^3 > 0. \quad (95)$$

These cases exhaust I , and therefore

$$D^3\Phi(x, 0)[r_1(x, 0), r_1(x, 0), r_1(x, 0)] > 0, \quad x \in I. \quad (96)$$

For the second contraction, equation (86) and $M\chi c^3 \geq 0$ give directly

$$\begin{aligned} D^3\Phi(x, 0)[r_2(x, 0), r_2(x, 0), r_2(x, 0)] &= (A_2 + c^3) + M\chi c^3 \\ &\geq \frac{1}{2} > 0, \quad x \in I. \end{aligned} \quad (97)$$

5.4. Passage to a convex open strip. The preceding argument gives strict inequalities only on the central segment. We now justify in detail that they persist on a convex open neighborhood. A direct differentiation of equation (67) gives the full Hessian

$$D^2\Phi(x, y) = \begin{pmatrix} \psi''(x) + \varepsilon g'(x)y + \frac{1}{6}b''(x)y^3 & \varepsilon g(x) + \frac{1}{2}b'(x)y^2 \\ \varepsilon g(x) + \frac{1}{2}b'(x)y^2 & L + b(x)y \end{pmatrix}. \quad (98)$$

In particular,

$$\Phi_{yy}(x, y) - \Phi_{xx}(x, y) = d(x) + (b(x) - \varepsilon g'(x))y - \frac{1}{6}b''(x)y^3. \quad (99)$$

The functions in equation (99) are smooth and bounded on a neighborhood of I . In view of equation (69), there is $\delta_1 > 0$ such that

$$\Phi_{yy}(x, y) - \Phi_{xx}(x, y) \geq \frac{d_0}{2} > 0 \quad (100)$$

for $(x, y) \in I \times [-\delta_1, \delta_1]$.

For such (x, y) , set

$$\mathfrak{d}(x, y) := \Phi_{yy}(x, y) - \Phi_{xx}(x, y), \quad \mathfrak{z}(x, y) := \Phi_{xy}(x, y), \quad (101)$$

$$\hat{\Delta}(x, y) := \sqrt{\mathfrak{d}(x, y)^2 + 4\mathfrak{z}(x, y)^2}, \quad \hat{\tau}(x, y) := -\frac{2\mathfrak{z}(x, y)}{\hat{\Delta}(x, y) + \mathfrak{d}(x, y)}, \quad (102)$$

and

$$\hat{c}(x, y) := (1 + \hat{\tau}(x, y)^2)^{-1/2}, \quad \hat{\sigma}(x, y) := \hat{\tau}(x, y)\hat{c}(x, y). \quad (103)$$

Then

$$\hat{r}_1(x, y) := (\hat{c}, \hat{\sigma}), \quad \hat{r}_2(x, y) := (-\hat{\sigma}, \hat{c}) \quad (104)$$

are smooth, consistently oriented unit eigenvectors of $D^2\Phi(x, y)$ for its lower and upper eigenvalues. Their eigenvalue gap is

$$\hat{\lambda}_2(x, y) - \hat{\lambda}_1(x, y) = \hat{\Delta}(x, y) \geq \mathfrak{d}(x, y) \geq \frac{d_0}{2}. \quad (105)$$

Thus the Hessian has two distinct real eigenvalues on the closed strip of height $2\delta_1$. At $y = 0$, the definitions in equations (102) and (104) reduce to equations (72) to (74); in particular, $\hat{r}_k(x, 0) = r_k(x, 0)$.

For $k = 1, 2$, define

$$\Psi_k(x, y) := D^3\Phi(x, y)[\hat{r}_k(x, y), \hat{r}_k(x, y), \hat{r}_k(x, y)]. \quad (106)$$

These functions are continuous. By equations (96) and (97), they are strictly positive on the compact segment $I \times \{0\}$. Hence

$$m_k := \min_{x \in I} \Psi_k(x, 0) > 0, \quad k = 1, 2. \quad (107)$$

Uniform continuity on a compact rectangle permits us to choose $0 < \delta \leq \delta_1$ so small that

$$\Psi_k(x, y) \geq \frac{m_k}{2} > 0, \quad (x, y) \in I \times [-\delta, \delta], \quad k = 1, 2. \quad (108)$$

We now fix this δ and take $\mathcal{V}$ as in equation (8). Since $DF = D^2\Phi$, equation (105) proves strict hyperbolicity of the flux $F = \nabla\Phi$ on $\mathcal{V}$. Combining lemma 2.4 with equation (108) gives

$$D\lambda_k(U)[\hat{r}_k(U)] = D^3\Phi(U)[\hat{r}_k(U), \hat{r}_k(U), \hat{r}_k(U)] > 0, \quad U \in \mathcal{V}, \quad k = 1, 2. \quad (109)$$

Thus both characteristic fields are genuinely nonlinear throughout the same convex open state domain.

6. ENTROPY COMPATIBILITY AND RECOVERY OF THE FOUR WELLS

With the choices above, define on $\mathcal{V}$

$$F := \nabla \Phi, \quad \eta(x, y) := \frac{x^2 + y^2}{2}, \quad q(x, y) := (x, y) \cdot \nabla \Phi(x, y) - \Phi(x, y). \quad (110)$$

Equivalently, the two components of the flux and the entropy flux are

$$F_1(x, y) = \psi'(x) + \varepsilon g(x)y + \frac{1}{6}b'(x)y^3, \quad (111)$$

$$F_2(x, y) = \varepsilon h(x) + Ly + \frac{1}{2}b(x)y^2, \quad (112)$$

$$q(x, y) = x\psi'(x) - \psi(x) + \varepsilon xg(x)y + \frac{L}{2}y^2 + \frac{1}{6}xb'(x)y^3 + \frac{1}{3}b(x)y^3. \quad (113)$$

Indeed, the first two identities follow by differentiating equation (67), and the third follows by substituting them into $q = (x, y) \cdot F - \Phi$. All three functions are smooth on a neighborhood of the closed rectangle $I \times [-\delta, \delta]$. By lemma 2.3,

$$D^2\eta = I_2 > 0, \quad Dq = D\eta DF \quad \text{on } \mathcal{V}. \quad (114)$$

At $U_i = (\xi_i, 0)$, equations (62) and (67) give

$$\Phi(U_i) = \psi(\xi_i), \quad \Phi_x(U_i) = \psi'(\xi_i), \quad \Phi_y(U_i) = \varepsilon h(\xi_i) = 0. \quad (115)$$

Therefore

$$F(U_i) = (\psi'(\xi_i), 0), \quad q(U_i) = \xi_i\psi'(\xi_i) - \psi(\xi_i). \quad (116)$$

Substituting the exact data from equation (51) yields

i	1	2	3	4
ξ_i	-7	-1	1	7
$-F_1(U_i)$	$-\frac{52}{5}$	$-\frac{34}{5}$	6	0
$-F_2(U_i)$	0	0	0	0
$\eta(U_i)$	$\frac{49}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{49}{2}$
$-q(U_i)$	$\frac{48}{5}$	$\frac{216}{5}$	-24	0.

(117)

Combining equations (5) and (117) gives, entry by entry,

$$G_{F,\eta,q}(U_i) = X_i, \quad i = 1, 2, 3, 4. \quad (118)$$

All four states lie in the interior of $\mathcal{V}$ because $\xi_i \in (-8, 8)$ and $0 \in (-\delta, \delta)$. It follows that

$$(X_1, X_2, X_3, X_4) \in \mathcal{K}_{F,\eta,q}^{(4)}. \quad (119)$$

The T_4 identities were verified in section 3, and equation (38) shows that the tuple is nondegenerate. Strict hyperbolicity and genuine nonlinearity were proved in equations (105) and (109), while equation (114) supplies the strictly convex entropy and its compatible entropy flux. This completes the proof of theorem 1.1. $\square$

APPENDIX A. EXACT-ARITHMETIC CHECKS

This appendix records compact checks of the numerical identities used in the construction. First, the rank-one factorizations and the closure condition can be checked entirely at the level of the two column vectors:

$$\begin{pmatrix} -26 \\ 0 \\ 24 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ -48 \end{pmatrix} + \begin{pmatrix} 6 \\ 0 \\ -24 \end{pmatrix} + \begin{pmatrix} 18 \\ 0 \\ 48 \end{pmatrix} = 0, \quad (120)$$

$$\begin{pmatrix} -26 \\ 0 \\ 24 \end{pmatrix} + \begin{pmatrix} -4 \\ 0 \\ 96 \end{pmatrix} + \begin{pmatrix} 30 \\ 0 \\ -120 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0, \quad (121)$$

after multiplying the matrices by 5. Second, the correction polynomial in equation (48) may be expanded as

$$xR(x)^2 = x^9 - 100x^7 + 2598x^5 - 4900x^3 + 2401x. \quad (122)$$

The vanishing of this polynomial and its first derivative at $x = \pm 1, \pm 7$ follows either from equation (47) or directly from equation (122). Substitution into equation (46) gives the eight rational values in equation (51); differentiating three times and substituting $x = -5, 0, 5$ gives exactly the three fractions in equation (52).

Finally, the lower-right entries in the four wells are recovered from

$$\psi(\xi_i) - \xi_i \psi'(\xi_i) = \left(\frac{48}{5}, \frac{216}{5}, -24, 0 \right)_i, \quad (123)$$

whereas their upper-right entries are

$$-\psi'(\xi_i) = \left(-\frac{52}{5}, -\frac{34}{5}, 6, 0 \right)_i. \quad (124)$$

These are precisely the second-column entries in equations (33) and (34).

APPENDIX B. THE SMOOTH CUTOFF FACT USED ABOVE

We used the following elementary one-dimensional form of the smooth cutoff lemma. If $K \subset \Omega \subset \mathbb{R}$, where K is compact and Ω is open, then there exists $\chi \in C_c^\infty(\Omega)$ such that $0 \leq \chi \leq 1$ and $\chi = 1$ on K . To see this, choose open sets O_1, O_2 satisfying

$$K \subset O_1, \quad \overline{O_1} \subset O_2, \quad \overline{O_2} \subset \Omega, \quad (125)$$

which is possible by compactness. Convolve the indicator function of O_2 with a nonnegative standard mollifier whose support is smaller than both $\text{dist}(\overline{O_1}, \mathbb{R} \setminus O_2)$ and $\text{dist}(\overline{O_2}, \mathbb{R} \setminus \Omega)$. The resulting smooth function is supported in Ω , takes values in $[0, 1]$, and equals one on O_1 , hence on K .